

ClassMic Business Case

B2 Investment decision for the course prototype

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Recommendation

GO for a two-sprint course prototype using existing equipment. We will test whether students can speak clearly through their phones while the instructor controls the queue. A funded classroom pilot remains at REVISE until we establish local demand, audio performance, support needs and the cost of commercial alternatives.

We request approval to develop and evaluate the prototype with an estimated 100 student hours. We plan no additional cash spending, assuming the team has enough time and the required equipment is available.

Opportunity and expected benefit

ClassMic aims to let the room hear student questions without waiting for a microphone to reach the speaker. Students join a temporary browser session, request a turn and transmit audio only after instructor approval. The instructor can mute a speaker or end the session.

The expected benefit is clearer questions with fewer pauses to repeat a question or pass a microphone. Local demand and performance still need testing, so we cannot yet estimate gains in participation or financial savings.

Basis for the decision

Basis	What it supports	What remains unknown
Proposed workflow	Phone microphones with instructor approval and stop controls.	How often students miss questions and whether they want this workflow.
Course scope	Five priority user stories built in two weekly sprints.	Approval, available resources and measured feasibility.
Biamp product information [1]	A commercial product offers the same core microphone and moderator workflow.	Local price, classroom fit and performance compared with the prototype.
Browser media specification [3]	Browser microphone capture is technically supported with permission controls.	Performance and compatibility on the devices and network we will use.

If we do nothing

Continuing current classroom practice avoids development and maintenance work but leaves any audibility problem unchanged. The prototype evaluation will compare ClassMic with that process.

Comparison of alternatives

The three alternatives address the same classroom objective. The assessment is qualitative because local costs, adoption and device performance remain unmeasured.

Criterion	Improve current process	Buy or adapt	Custom ClassMic prototype
Strategic fit	Support discussion by repeating questions or using an existing microphone.	Existing audience microphone workflow may fit the classroom.	Focused test of browser access and instructor control.
Expected benefit	May solve the problem with a small change in teaching practice.	Phone speaking and moderation are established product capabilities [1].	Test audibility and ease of use against the current speaking process.
Cost and value	Uses existing teaching practices and equipment. Staff time still matters.	Room hardware and installation require a quote [2]. Total cost unknown.	Estimated 100 student hours and AED 0 additional cash. Long-term maintenance is outside the estimate.
Feasibility	Lowest technical complexity if the existing workaround is adequate.	Needs procurement, AV integration and a product trial.	Browser, network, feedback and permission behavior need testing.
Schedule	Can begin after agreement on the classroom routine.	Delivery and installation dates require a supplier response.	Two weekly sprints are planned. Audio is the main schedule risk.
Security and privacy	No new microphone software or data flow.	Review actual configuration, data handling and support access.	Require microphone permission and instructor control. Store no audio.
Operational support	Uses current teaching and AV support arrangements.	Supplier and institution divide support duties. Terms are unknown.	Two students support the prototype. Production support is outside scope.
Stakeholder adoption	Familiar process, but repeated questions may remain disruptive.	Test the attendee and moderator workflow with real users.	Test whether students can join and use the browser workflow easily.

Reason for the recommendation

Building a prototype meets the course requirement to develop and evaluate selected user stories. A later campus decision should also compare a commercial product with improvements to current classroom practice.

Proposed delivery and resources

Stage	Planned dates	Decision or output
Initiation and planning	10–18 September	Project scope, requirements and schedule reviewed before development.
Sprint 1	19–25 September	Session access, microphone permissions, speaking queue and instructor controls.
Sprint 2	26 September–2 October	Audio integration, misuse and recovery tests, volunteer feedback and the product report.
Closing	3–9 October	Assess results against the project objectives and prepare the final recommendation.

The proposed plan review is on 18 September, followed by two weekly sprints. The planning assignment is due 22 September at 00:00. Development, testing and the product report must finish by 2 October for the report deadline of 3 October at 00:00. Execution evidence is due 4 October at 00:00 and closing work on 10 October at 00:00.

Preliminary cost and effort

Work	Temiko hours	Nurtore hours	Total hours
Initiation and planning	18	22	40
Sprint 1	18	6	24
Sprint 2	14	10	24
Closing and presentation	4	8	12
Total hours	54	46	100

The AED 0 cash estimate assumes we can reuse available phones, a laptop, speakers and permitted tools. The 100 student hours are an effort estimate; no hourly pay rate has been assigned. Paid hosting, equipment or account upgrades would require a revised budget and approval.

Feasibility conditions

Feasibility depends on access to at least two suitable phones, a laptop and a speaker. Early tests will assess microphone permissions and browser behavior. Evaluation will use volunteers in a controlled setting. Room access, advisers and participants remain unconfirmed.

We will measure one-way audio delay in 30 trials, aiming for a 95th percentile of 150 ms or less. Tests must also check whether speech is intelligible and whether speaker feedback occurs.

Decision gates and principal risks

Gate	Evidence required	Decision
Before Sprint 1	Team and instructor review, five-story scope, resources, test plan and proposed schedule.	GO with the prototype only if the plan is feasible. Otherwise revise scope or dates.
After Sprint 1	Session and queue demo, microphone feasibility check, feedback and updated backlog.	Continue core audio integration. Drop optional work if capacity or compatibility is weak.
End of Sprint 2	All required checks must pass, with no open critical defect. At least 9/10 setups within 2 minutes, 27/30 speaking trials within 3 seconds and 75% positive feedback from at least 5 reviewers.	Complete the course evaluation. Report unmet targets and test limitations.
Any future live pilot	Local need, acceptable audio, named support owner, privacy and access review, supplier comparison and funded estimate.	REVISE until evidence is sufficient. Seek a separate authorization.

Risks that could change the decision

Speaker feedback or long network delays could make speech hard to follow. Some phones or browsers may be incompatible, and weak instructor controls could allow unwanted transmission. Limited team time or unavailable reviewers could reduce testing. Stop tests if control or privacy fails, keep a physical microphone or instructor repetition as a fallback, and reduce optional work if time is short.

Investment conclusion

Proceed with the course prototype using available equipment and an estimated 100 student hours. Consider a larger pilot only if the tests show clearer questions, reliable instructor control and a feasible support plan.

Sources

[1] [Biamp Crowd Mics product overview](#)

[2] [Biamp Crowd Mics FAQ, including quote requirements and audio latency](#)

[3] [W3C Media Capture and Streams specification](#)

Sources accessed 17 September 2026.